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Sampling from an Urn: With and Without Replacement

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1 Introduction and experimental design

An urn contains 117 white balls and 191 black balls, so the total number of balls is

$$N = 117 + 191 = 308, \qquad p = P(\text{white}) = \frac{117}{308} = 0.379870, \qquad q = 1 - p = 0.620130.$$

Two experiments were studied. In the first experiment, a ball was returned to the urn after every draw. In the second experiment, a selected ball was not returned. Each experiment consisted of 20 draws, and each model was simulated 500 times in the supplied workbook. Thus, across all 500 experiments, a total of 10,000 draws were recorded.

The data-generation sheets contain volatile random formulas. One generated realization was copied as fixed values into the computation sheets before the numerical analysis was performed. Consequently, the results in this report remain unchanged when the workbook recalculates. White and black outcomes are represented by W and B, respectively.

For a theoretical quantity t and a comparison value a, relative error is reported as

$$\mathrm{RE}(a;t) = \frac{|a-t|}{|t|} \times 100\%.$$

Let $\hat{f}(x)$ denote the simulated relative frequency of drawing exactly x white balls across the 500 experiments. The simulated frequencies need not equal the theoretical probabilities exactly. Sampling variation is expected because only 500 repetitions were performed, and rare events may occur very few times or not at all.

2 Drawing With Replacement

2.1 Overall Proportion

With replacement, the 20 draws in one experiment are independent Bernoulli trials with a constant success probability p, where success means drawing a white ball. Across the 10,000 simulated draws, the observed white proportion was 0.378400, compared with the theoretical value 0.379870. The absolute deviation was 0.001470 and the relative error was 0.3870%. This close agreement is consistent with the law of large numbers, which guarantees that the sample proportion converges to the true probability as the number of draws increases.

The small discrepancy shows that the simulated draws reproduce the urn's white-ball proportion well. The remaining difference is expected random variation rather than a systematic change in the probability.

2.2 Binomial Model

Let X be the number of white balls in one 20-draw experiment. It therefore has the binomial distribution

$$P(X = x) = \binom{20}{x} p^x q^{20-x}, \qquad x = 0, 1, \ldots, 20.$$

Table 1 compares the binomial probabilities with the relative frequencies from the 500 experiments. Both columns sum to 1, up to rounding. The largest discrepancies occur at individual values such as $x = 10$, but the overall shape remains centred around 7 or 8 white balls.

Table 1: Binomial probabilities and simulated relative frequencies.

x	$P(X=x)$	$\hat{f}(x)$	x	$P(X=x)$	$\hat{f}(x)$
0	0.000071	0.000	11	0.054142	0.062
1	0.000867	0.000	12	0.024874	0.034
2	0.005043	0.008	13	0.009377	0.010
3	0.018536	0.018	14	0.002872	0.004
4	0.048257	0.036	15	0.000704	0.000
5	0.094593	0.106	16	0.000135	0.000
6	0.144861	0.164	17	0.000019	0.000
7	0.177474	0.178	18	0.000002	0.000
8	0.176661	0.158	19	0.000000	0.000
9	0.144288	0.166	20	0.000000	0.000
10	0.097225	0.056			

Most of the probability is concentrated around six to nine white balls, while counts near the two ends are very rare. Therefore, zero simulated frequencies at some extreme values are reasonable with only 500 experiments.

2.3 Expected Value

For a binomial random variable, $E(X) = 20p$. Here,

$$E(X) = 20\left(\frac{117}{308}\right) = 7.597403.$$

The simulated mean was 7.568000, an absolute deviation of 0.029403 and a relative error of 0.3870%. The sample mean is therefore very close to its theoretical target.

This agreement confirms that the simulation captures the long-run number of white balls per experiment, not only the overall proportion of individual white draws.

2.4 Short Initial Sequences

The event of no white ball among the first three draws is the event BBB. Independence gives

$$P(BBB) = q^3 = 0.238478.$$

The empirical frequency was 0.228000. Its absolute deviation was 0.010478 and its relative error was 4.3936%. The larger percentage error, compared with the overall white proportion, is reasonable because this calculation uses only 500 three-draw sequences rather than 10,000 individual draws.

Although the relative error is larger, the absolute difference is only about 0.0105. This illustrates why relative errors can appear noticeable for events estimated from a smaller number of observations.

2.5 Four white balls before five black balls

Suppose the fourth white ball arrives after exactly b black balls, where $b = 0, 1, 2, 3, 4$. The last draw must be white, and among the preceding $b + 3$ positions there must be exactly three white balls. Therefore,

$$P(4W \text{ before } 5B) = \sum_{b=0}^{4} \binom{b+3}{3} p^4 q^b = 0.359647.$$

Equivalently, after the first eight draws at least four outcomes must be white; by that time either the fourth white or the fifth black must already have appeared. The simulated relative frequency was 0.372000, giving an absolute deviation of 0.012353 and a relative error of 3.4347%.

The simulated value is close to the theoretical result, supporting the counting argument based on the possible numbers of black balls appearing before the fourth white ball.

2.6 Approximating the lower binomial tail

The event "less than six white balls" is $X \leq 5$. Its exact binomial probability is

$$P(X < 6) = \sum_{x=0}^{5} \binom{20}{x} p^x q^{20-x} = 0.167366.$$

The empirical frequency was 0.168000. Its relative error from the exact value was only 0.3787%.

For the Poisson approximation, $\lambda = np = 7.597403$, so

$$P(X < 6) \approx \sum_{x=0}^{5} e^{-\lambda} \frac{\lambda^x}{x!} = 0.230956.$$

This differs from the exact binomial value by 0.063589, a relative error of 37.9941%. The approximation is poor because $p \approx 0.38$ is not small, whereas the Poisson approximation is most reliable when the number of trials is large and the success probability is small.

For the normal approximation, the de Moivre–Laplace central limit theorem justifies treating the binomial distribution as approximately normal when n is sufficiently large. Here,

$$\mu = np = 7.597403, \qquad \sigma = \sqrt{npq} = \sqrt{4.711376} = 2.170571.$$

The continuity-corrected boundary for $X \leq 5$ is 5.5. Excel gives

$$P(X < 6) \approx \Phi\left(\frac{5.5 - 7.597403}{2.170571}\right) = \Phi(-0.966291) = 0.166949.$$

The error relative to the exact binomial value is 0.2491%. For a printed standard-normal table, the standardized value is rounded to -0.97, giving $\Phi(-0.97) = 0.1660$. This table value differs from Excel's unrounded normal calculation by 0.5686%, and from the exact binomial probability by 0.8163%. The normal approximation is substantially better than the Poisson approximation in this case.

Table 2: Comparison of methods for $P(X < 6)$.

Method	Probability	Relative error from exact binomial
Exact binomial	0.167366	–
Poisson approximation	0.230956	37.9941%
Normal CDF, continuity corrected	0.166949	0.2491%
Standard-normal table, $z = -0.97$	0.1660	0.8163%
Simulation	0.168000	0.3787%

Overall, the continuity-corrected normal model gives the best approximation because both (np) and (nq) are reasonably large. The comparison also shows that an approximation should be chosen according to its assumptions, not merely because it is easy to calculate.

3 Drawing without replacement

3.1 Conditional probabilities and the hypergeometric law

Without replacement, successive draws are dependent. If m_1 white balls and n_1 black balls have already been drawn, the conditional probability that the next ball is white is

$$P(\text{next is white} \mid m_1, n_1) = \frac{117 - m_1}{308 - m_1 - n_1}.$$

The simulation updates this probability after every draw. The total number X of white balls in 20 draws follows a hypergeometric distribution:

$$P(X = x) = \frac{\binom{117}{x}\binom{191}{20-x}}{\binom{308}{20}}, \qquad x = 0, 1, \ldots, 20.$$

Table 3 compares the hypergeometric probabilities with the simulated relative frequencies.

Table 3: Hypergeometric probabilities and simulated relative frequencies.

x	$P(X=x)$	$\hat{f}(x)$	x	$P(X=x)$	$\hat{f}(x)$
0	0.000047	0.000	11	0.051717	0.050
1	0.000645	0.002	12	0.022467	0.032
2	0.004109	0.004	13	0.007890	0.006
3	0.016293	0.020	14	0.002218	0.006
4	0.045108	0.048	15	0.000491	0.002
5	0.092677	0.092	16	0.000084	0.000
6	0.146608	0.146	17	0.000011	0.000
7	0.182849	0.178	18	0.000001	0.000
8	0.182593	0.178	19	0.000000	0.000
9	0.147427	0.148	20	0.000000	0.000
10	0.096764	0.088			

The simulated frequencies follow the central shape of the hypergeometric distribution, with the largest probabilities again near seven and eight white balls. Small differences at individual values are expected from the finite number of experiments.

3.2 Expected value

Label the 117 white balls and define $I_j = 1$ if white ball j is included in the 20-ball sample, and $I_j = 0$ otherwise. Then $X = \sum_{j=1}^{117} I_j$. By symmetry, any particular ball is included with probability 20/308. Linearity of expectation does not require independence, so

$$E[X] = \sum_{j=1}^{117} E[I_j] = 117\left(\frac{20}{308}\right) = 20\left(\frac{117}{308}\right) = 7.597403.$$

The simulated mean without replacement was 7.606000. Its absolute deviation from 7.597403 was 0.008597, or 0.1132%. Thus, the simulated mean is close to the theoretical value, just as in the with-replacement experiment.

The unchanged theoretical mean shows that removing balls affects dependence and variability, but not the average number of white balls in a sample of fixed size.

3.3 No white ball among the first three draws

The probability that the first three balls are all black is now

$$\frac{191}{308}\frac{190}{307}\frac{189}{306} = \frac{\binom{191}{3}}{\binom{308}{3}} = 0.237049.$$

The empirical frequency was 0.246000, producing an absolute deviation of 0.008951 and a relative error of 3.7760%.

The theoretical probability is slightly lower than in the with-replacement case because each removed black ball makes the next black draw less likely. The simulated ordering is reversed in this realization, which is consistent with ordinary sampling variation.

4 Comparison of the two sampling models

The two models have the same one-draw probability $p = 117/308$ and the same expected number of white balls, 7.597403. Their dependence structures differ. With replacement, the binomial variance is

$$20pq = 4.711376.$$

Without replacement, the finite-population correction reduces the variance to

$$20pq\frac{308-20}{308-1} = 4.419793.$$

Thus, the hypergeometric distribution is slightly more concentrated near its mean. This can be seen in the probabilities for $x = 7$ and $x = 8$, which are both larger without replacement than under the binomial model.

The probability of three initial black balls also decreases from 0.238478 with replacement to 0.237049 without replacement. After each black ball is removed, the remaining proportion of black balls falls, making another black draw slightly less likely. The theoretical difference is small because the sample of three balls is small compared with the population of 308 balls.

Table 4: Summary comparison of theoretical and simulated quantities.

Quantity	Theory, repl.	Sim., repl.	Theory, no repl.	Sim., no repl.
Mean white count	7.597403	7.568000	7.597403	7.606000
No white in first 3	0.238478	0.228000	0.237049	0.246000
Variance of white count	4.711376	–	4.419793	–

5 Conclusion

The simulations agree well with the principal theoretical predictions. The overall white proportion and both sample means are close to their expected values, and the binomial and hypergeometric probability tables have the anticipated shapes. Differences between theory and simulation are attributable to ordinary sampling variation in 500 experiments.

For the lower binomial tail, the continuity-corrected normal approximation was accurate, whereas the Poisson approximation was not appropriate for the relatively large success probability. Removing balls creates dependence and slightly reduces the variance through the finite-population correction, but it does not change the expected white count. These observations illustrate how the sampling mechanism affects a distribution even when the one-draw proportion remains the same.

Contribution

The project work was divided so that each member contributed to the modelling, computation, writing, and checking stages. The final report and workbook were reviewed together to keep the notation, numerical values, and explanations consistent.

Student	Student ID	Contribution
Zhang Yang	MAT2509036	Coordinated the report structure, wrote the introduction and experimental design, and developed the with-replacement binomial model, including the overall white proportion, the expected value, and the first-three-draw event.
Zhu Xinbo	MAT2509040	Prepared and checked the Excel simulations for the with-replacement case, fixed the generated data as values, computed the relative-frequency tables, and verified the event that four white balls occur before five black balls.
Chen Yanfu	MAT2509003	Developed the without-replacement model, including the conditional probability formula, the hypergeometric probability mass function, the expectation proof, and the comparison between theoretical and simulated values.
Zhang Qixuan	MAT2509034	Completed the approximation and comparison sections, including the Poisson and normal approximations, relative-error calculations, variance comparison, conclusion, LaTeX formatting, and final proofreading.